Hiba Amin

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Lahore, Pakistan
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Profile

To obtain a position in a dynamic software house where I can apply my knowledge of programming, web/app development, and problem-solving to contribute to real-world projects, while continuously learning and growing in a professional environment.

Education

Mphil Computer Science, *University of Punjab*
2024 – present
CGPA:3.7/4 Secured 2nd position in my class

Bs Computer Science,
Lahore College for women university
2020 – 2024
CGPA: 3.60/4

Skills

Python
Flask, OpenCV, Matplotlib, Seaborn, Pandas, Numpy,scikit learn, tensor flow

Ionic Framework

Databases
MySQL, SQL Lite

JavaScript
Vanilla JS

Version Control
Git, Github

Certificates

Freelancing (Digi-skills)
Learning online work and client communication

Digital Marketing (Digi-skills)
Learning online marketing. social media, and SEO basics.

Google Project Management
Gaining skills in planning, teamwork, and project management techniques

Data Analytics with Python. (IBM)
Learning data cleaning, analysis, visualization, and building machine learning models

Professional Experience

Data Science Intern, *10Pearls*
06/2025 – 08/2025 | Lahore
Worked as a Data Science Intern at 10Pearls, where I built an AQI prediction model using Linear Regression, Random Forest, and XGBoost. Developed a Streamlit dashboard, integrated Hopsworks feature store, and implemented CI/CD pipeline for real-time air quality monitoring.

Projects

AQI Predictor – Data Science Intern (10Pearls)
06/2025 – Present
- Working on a real-time Air Quality Index (AQI) prediction system using Python.
- Using hopswork to store feature group.
- Applied data cleaning, feature engineering, and trained ML models (e.g., Linear Regression, XGBoost, LSTM) for air quality forecasting.
- Integrated live data collection using public APIs (e.g., AirVisual, OpenWeatherMap).
- Visualized trends and model performance using Matplotlib and Seaborn.
- Improved model accuracy and deployed pipeline using Github Action for real-time analysis.

PULMO-X: An Android based Application for the detection of Pulmonary Diseases using Chest X-ray images (Final year)
- Developed PULMO-X, an Android application aimed at the early detection of pulmonary diseases including Pneumonia, Tuberculosis (TB), and Covid-19.
- Utilized Artificial Intelligence (AI) and Machine Learning (ML) models, implemented in Python, for accurate analysis of chest X-ray images.
- Enabled users to upload chest X-rays through the app for instant, automated diagnostic feedback.
- Designed a user-friendly interface for smooth interaction, image upload, and result display.
- Focused on early detection and accessibility, especially for use in remote or resource-limited areas.
- Final year project demonstrating cross-platform integration of mobile development and AI-based medical diagnostics.

Expert Systems Project: Fever Diagnosis and Recommendation System
- Developed an expert system in SWI-Prolog for diagnosing fever-related diseases such as dengue, malaria, and typhoid.
- Utilized rule-based reasoning to infer possible illnesses based on user-reported symptoms.
- Designed an interactive user interface to allow users to input symptoms and receive a diagnosis.
- Implemented a knowledge base of symptoms and disease rules to enhance decision accuracy.
- System provides medicine recommendations tailored to the diagnosed illness.

Course Management System
- Developed a Course Management System using MySQL to centralize and manage departmental academic data.
- Implemented functionalities to track semester-wise course enrollments and offerings.
- Enabled management of faculty assignments to different courses each semester.
- Maintained detailed student enrollment statistics for academic analysis and reporting.
- Designed database schema to ensure data consistency, integrity, and scalability.
- Facilitated efficient academic planning through streamlined data access and updates.

Fire and Gas Detection System
- Contributed to the design and documentation of a Fire and Gas Detection System for industrial safety applications.
- Focused on creating a detailed Software Requirements Specification (SRS) document outlining system functionality and performance needs.
- Developed UML models including use case diagrams, class diagrams, and sequence diagrams to visualize system architecture and behavior.
- Collaborated in defining functional and non-functional requirements for accurate and timely hazard detection.
- Aimed to ensure system reliability, scalability, and real-time responsiveness through structured documentation.
- Supported the groundwork for future software development and safety compliance.

Web site Development
- Developed a visually appealing website to showcase various types of paintings using HTML, CSS, and JavaScript.
- Designed user-friendly layouts for smooth navigation and engaging user experience.
- Displayed artwork and artist details with a focus on clarity, accessibility, and visual balance.
- Implemented responsive design to ensure compatibility across devices (desktop, tablet, mobile).
- Focused on aesthetic appeal to enhance visitor engagement and browsing satisfaction.
- Applied JavaScript for dynamic content rendering and interactivity (e.g., image previews, filters, or modals).